EXPERIMENT – 2

NAME : D.SARVESHVARAN ROLL NO : 240701480

Design a UI where users recall visual elements (e.g., icons or text chunks). Evaluate the effect of chunking on user memory.

FRAME 1:

INSTRUCTION PAGE:

Chunking Analysis of the Instruction Page Chunking is a cognitive strategy that breaks down information into smaller, manageable units, making it easier to process and retain. The Memory Recall Task instruction page effectively utilizes chunking in the following ways:

1.Clear and Sequential Numbering

- Instructions are presented as numbered steps.
- Each step focuses on a single action, reducing cognitive overload.
- Users can easily understand the flow of the task.

2. Logical Grouping of Information

- **Observation Phase** (step-1)– Explains what users will see.
- **Memorization Phase**(step-2) – Encourages users to remember items.
- **Recall Phase** (step-3)– Guides users on how to recall information.
- **Performance Criteria**(step-4) – Emphasizes accuracy over speed.

3.Visual Hierarchy and Design Elements

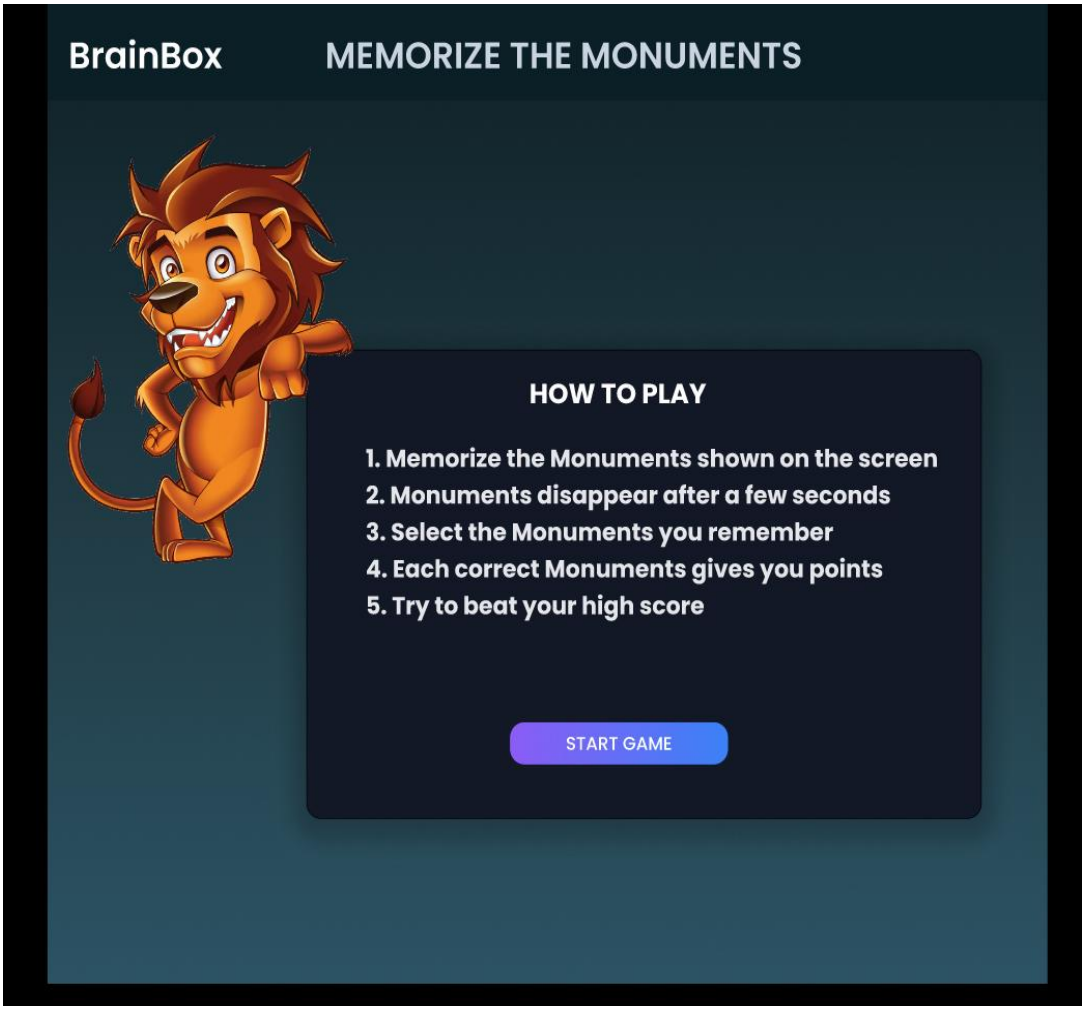
- Bold title("**BrainBox– Memorize Monuments**") clearly indicates the task.
- Adequate spacing improves readability.
- Highlighted **START** button guides users to the next step.
- Mascot Themed visuals make the interface engaging and playful without distraction.

4. Time Constraint Reinforcement(5 seconds)

- Clearly mentions the limited viewing time.
- Sets user expectations and improves focus.

5. Simplicity and Clarity

- Instructions use short and simple sentences.
- Active voice makes the task easy to understand and follow.



FRAME 2:

CHUNKING PHASE:

Analysis of the Memory Recall Task - Chunking Phase Screen

This screen represents the Chunking Phase of a Memory Recall Task, where users observe and memorize different monuments within a limited time. Below is a breakdown of its key components:

1. Purpose of the Screen

- Allows users to view multiple monuments within a fixed time.
- Encourages users to group similar monuments using chunking strategies.
- Helps convert individual monuments into meaningful clusters.

2. Key UI Elements

- **Countdown Timer**
    - Displays remaining viewing time.
    - Creates urgency and maintains focus.
- **Progress Bar**
    - Visually represents time passage.
    - Reinforces awareness of limited duration.
- **Grid of Icons**
    - monuments are arranged in a structured grid.
    - Users can chunk items based on:
        - Category (historical,temples,picnic spots)
        - Colour similarities
        - places
- **Mascot Cartoon**
    - Adds a playful and friendly tone.
    - Improves engagement and motivation.

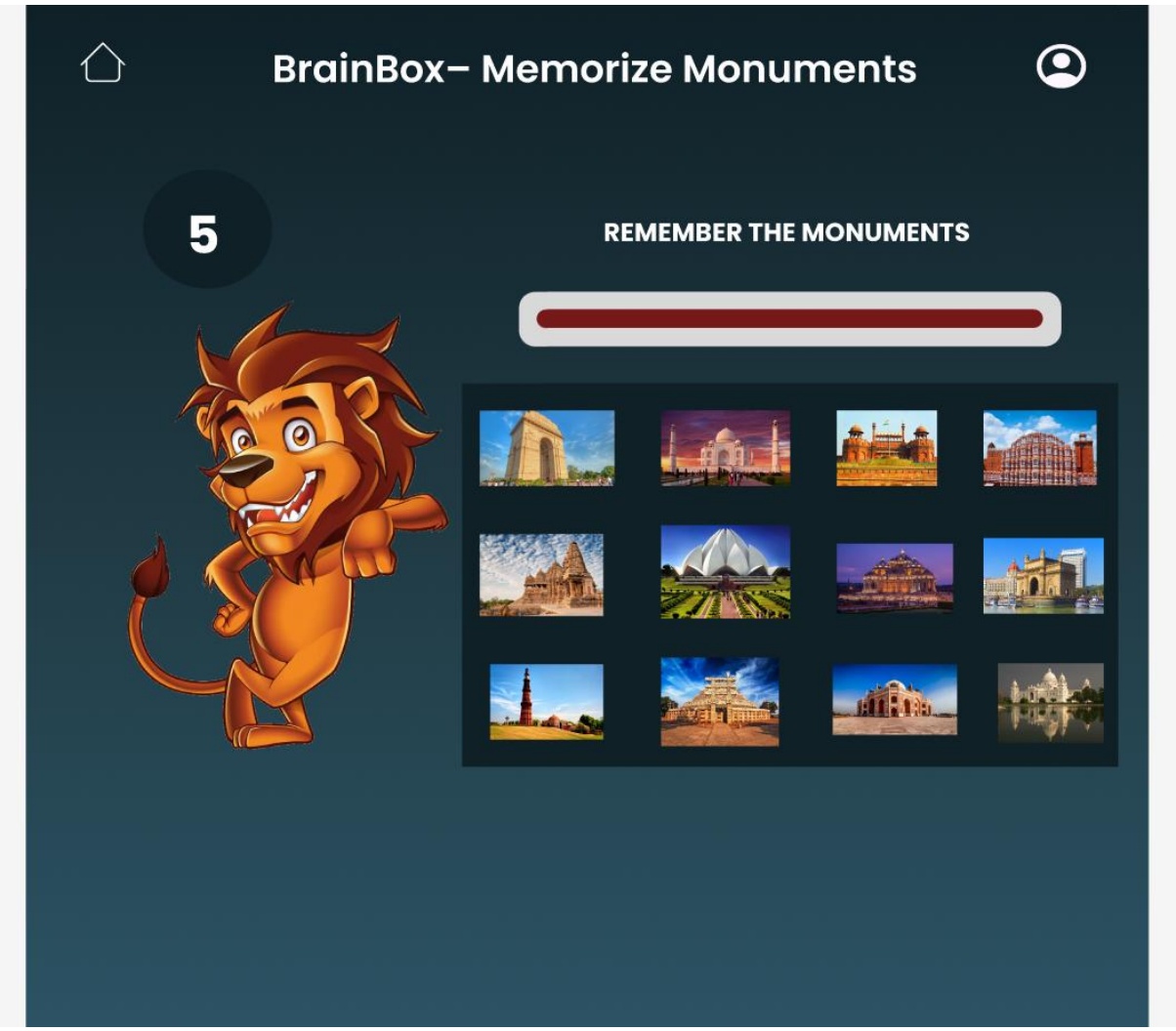
3. How the Chunking Phase Works

1. Users scan the grid.
2. Identify the monuments and store them in their mind.
3. Prepare for recall once time ends.

4. Cognitive Benefits

- Reduces memory load.
- Improves image recognition.

- Enhances short-term memory retention.



FRAME 3:

RECALL PHASE:

Analysis of the Memory Recall Task - Selection Phase

This screen represents the memory retrieval stage, where users select monuments they remember from the chunking phase.

1. Purpose of the Screen

- Tests how effectively users retained information.
- Measures recall accuracy after chunking.

2. Key UI Components

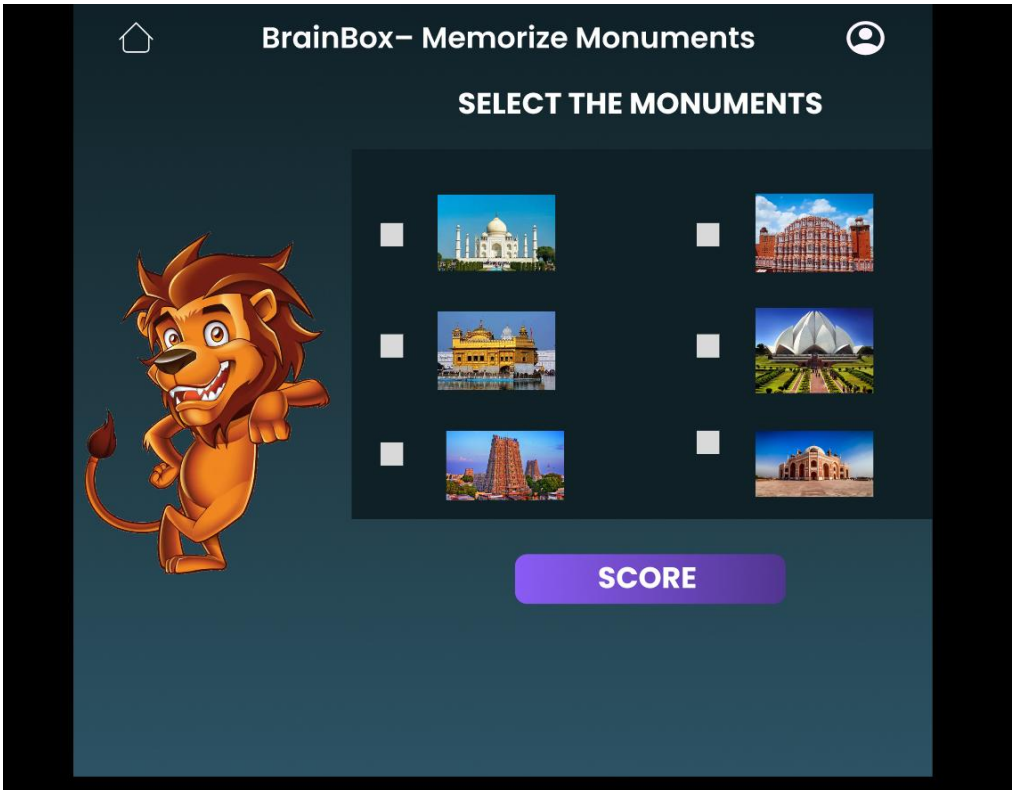
- Title and Instructions
  - Reinforces the task objective.
  - Clearly instructs users to select remembered monuments.
- Grid of Icon Choices
  - Includes both correct monuments and distractors.
  - Users must differentiate between seen and unseen monuments .
  - Selection controls (76checkboxes) make interaction simple.
- Submit Button
  - Confirms user choices.
  - Moves the task to evaluation stage.

3. How the Recall Phase Works

1. Users recall previously seen monuments.
2. Select matching monuments from options.
3. Avoid distractor monuments.
4. Submit responses for evaluation.

4. Cognitive & UX Benefits

- Tests memory accuracy.
- Evaluates false memory recognition.
- Keeps interaction intuitive and engaging.



FRAME 4:

RESULT PAGE

Analysis of the Memory Recall Task – Score & Feedback Screen

This screen provides performance feedback based on the recall phase.

1. Purpose of the Screen

- Displays recall accuracy.
- Allows users to take further action.

2. Key UI Components

- **Score Display**
  - Shows correct answers out of total attempts.
  - Makes performance easy to understand.
- **Action Buttons**
  - **Try Again –** Retry the task.
  - **Exit –** End the game.
- **Visual Elements**

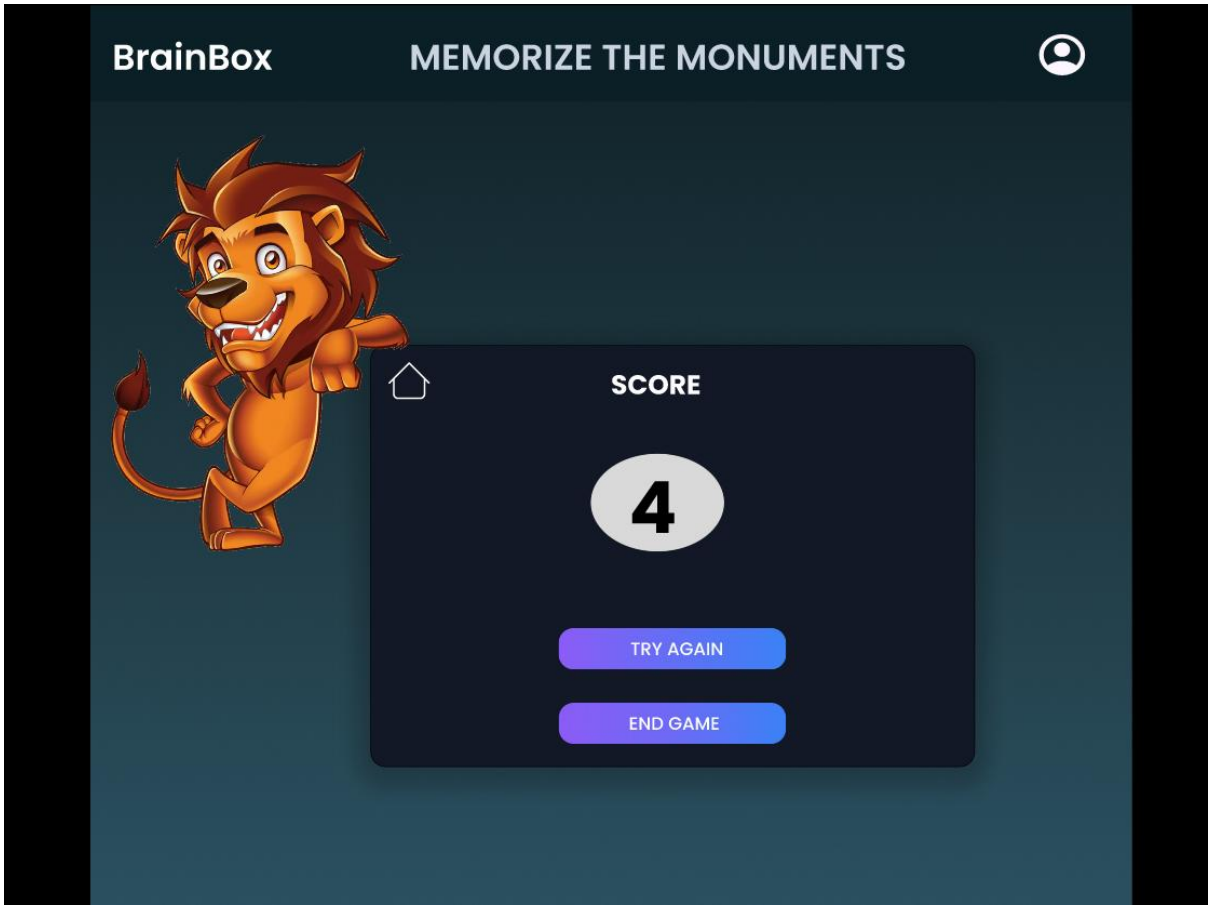
- Mascot and illustrations reduce stress.
- Gamified elements improve user experience.

3. How This Phase Works

1. System evaluates user selections.
2. Calculates recall score.
3. Displays feedback visually.
4. Users choose their next action.

4. Cognitive & UX Benefits

- Immediate feedback reinforces learning.
- User control improves engagement.
- Gamification keeps the task enjoyable.



PROTYPE LINK:

https://www.figma.com/design/O7ETo6N94kBPGHnLf9i4XP/Untitled?node-id=0-1&t=KMgd1cPLL3IgaacE-1